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Acceptability of short term neo-adjuvant androgen deprivation in patients with locally advanced prostate cancer

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Abstract

Purpose: To determine the acceptability of short term neo-adjuvant maximal androgen deprivation (MAD) to patients treated with external beam radiation for locally advanced prostate cancer.

Methods: Between 1996 and 2000, 818 patients with locally advanced, but non-metastatic, prostate cancer were entered into a randomised clinical trial (TROG 96.01), which compared radiation treatment alone with the same radiation treatment and 3 or 6 months neo-adjuvant MAD with goserelin and flutamide. Relevant symptoms, and how troublesome they were to the patient, were scored using a self-assessment questionnaire. This was completed by the patient at registration, and at specified times during and after treatment. Patients taking flutamide had liver function tests checked at regular intervals.

Results: All patients have completed at least 12 months follow-up after treatment. Nearly all patients completed planned treatment with goserelin, but 27% of patients in the 6-month MAD treatment arm, and 20% in the 3-month arm, had to stop flutamide early. This was mainly due to altered liver function (up to 17% patients) and bowel side effects (up to 8% patients). However, although flutamide resulted in more bowel symptoms for patients on MAD, there was significant reduction in some urinary symptoms on this treatment. Acute bowel and urinary side effects at the end of radiation treatment were similar in all treatment arms. Side effect severity was unrelated to radiation target volume size, which was reduced by MAD, but symptomatology prior to any treatment was a powerful predictor. Of the 36% of patients who were sexually active before any treatment, the majority became inactive whilst on MAD. However, sexual activity at 12 months after radiation treatment was similar in all treatment arms, indicating that the effects of short term MAD on sexual function are reversible.

Conclusion: Despite temporary effects on sexual activity, and compliance difficulties with flutamide, short-term neo-adjuvant MAD was not perceived by patients to be a major inconvenience. If neo-adjuvant MAD in the way tested can be demonstrated to lead to improved biochemical control and/or survival, then patients would view these therapeutic gains as worthwhile. Compliance with short-term goserelin was excellent, confirming that LH–RH analogues have a potential role in more long-term adjuvant treatment. However, for more protracted androgen deprivation, survival advantages and deleterious effects need to be assessed in parallel, in order to determine the optimal duration of treatment.
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Keywords: Prostate cancer; External beam radiation treatment; Neo-adjuvant androgen deprivation; Side-effects

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1. Introduction

In 1996, the Trans-Tasman Radiation Oncology Group initiated a randomised clinical trial (96.01) investigating the effectiveness of different durations of maximal androgen deprivation (MAD) using goserelin (Registered tradename: Zoladex, AstraZeneca Pharmaceuticals) and flutamide (Registered tradename: Eulexin, Schering-Plough Pharmaceuticals) prior to and during definitive radiation therapy for locally advanced carcinoma of the prostate. The principal objective of the trial was to determine whether initial treatment with MAD improved local and distant control, and if so, whether the magnitude of benefit correlated with the duration of MAD. One of the secondary objectives was to see how the addition of MAD affected toxicity profiles.

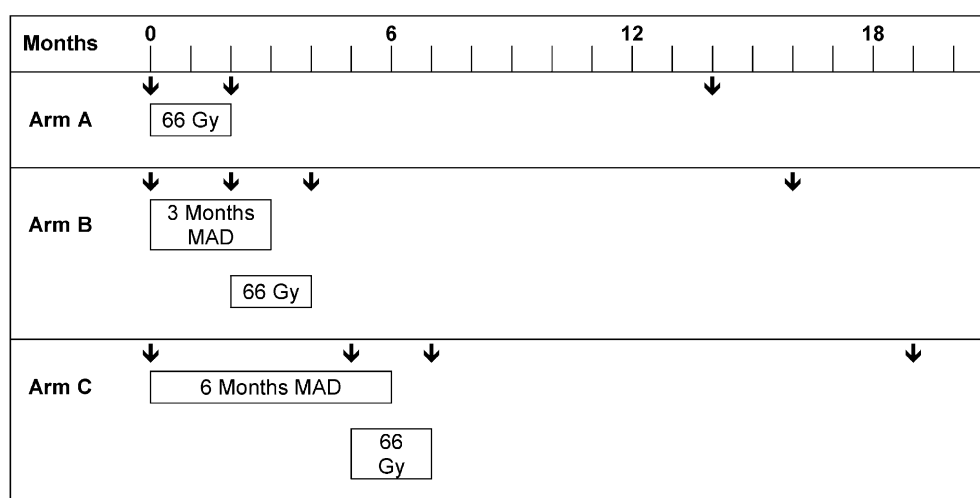
Fig. 1 shows the schema for the trial. All patients had locally advanced adenocarcinoma of the prostate (T2b or more—UICC 1992), but no radiological evidence of lymph node or distant metastases. Patients were randomised to radiotherapy alone, or 3 months MAD (beginning 2 months before radiotherapy), or 6 months MAD (beginning 5 months before radiotherapy). Randomisation was balanced by stratification according to stage, grade (Gleason score), presenting PSA and age. An important component of this trial was the completion by each patient of a detailed self-assessment questionnaire (SAQ). This scored bowel and urinary symptoms, and the inconvenience they caused, as well as assessing if sexual activity was possible. The SAQ was completed at randomisation, prior to radiotherapy, and at specified intervals after radiotherapy.

The immediate toxicity experienced by patients treated in this trial is important for a number of reasons. Evidence now exists that substantial benefits can be achieved with prolonged adjuvant androgen deprivation for selected

patients with localised prostate cancer [3,4,7,12], so the toxicity of, and compliance with, the various medications that can be used in this situation need to be defined. It is also important to be certain that medications delivered just before, and concurrently with, definitive radiation treatment do not exaggerate any of the acute (or late) side effects of radiotherapy. Conversely, neo-adjuvant androgen deprivation could lead to a reduction in acute radiation side effects, if a reduction in the prostatic volume prior to radiation leads to a reduction in the radiation target volume. The incidence and severity of delayed radiation side effects, particularly rectal injury, are known to correlate positively with increasing size of the radiation target volume [16], but the influence of target volume size on acute radiation side effects is less clear.

Radiation treatment alone results in 18–27% of patients becoming impotent within the first 12–15 months after treatment [1], but erectile dysfunction increases with advancing time, and 39–70% are impotent by 5 years after treatment [17]. Sexual activity before and after definitive radiation treatment for prostate cancer in Australasian men has been previously reported [22], but the additional use of MAD prior to radiotherapy can be expected to compound the problem of impotence whilst the patient is taking the medication, and for a while afterwards. The impact of MAD on ultimate recovery of sexual function after the combined treatment is not known. It is also uncertain whether younger patients experience less disturbance of sexual function compared to older patients.

This report attempts to address some of these previously unanswered questions. In addition, the inconvenience to the patient of taking short-term neo-adjuvant MAD is assessed, so the likely acceptability of more protracted treatment can be predicted.



↓ Timing of SAQs analysed in this report.

Fig. 1. Schematic representation of the treatment and timing of the SAQs.

2. Methods

MAD was achieved using a combination of the LH–RH analogue goserelin 3.6 mg subcutaneously once monthly, together with the non-steroidal anti-androgen flutamide 750 mg daily by mouth. Investigators followed a protocol of dose reduction in response to toxicity from flutamide. In the event of gastrointestinal side effects unacceptable to the patient, the dose of flutamide was initially reduced from 750 to 500 mg daily. If the gastrointestinal side effects improved on the reduced dosage, patients were maintained on the lower dose for the remainder of the allocated period of MAD. If gastrointestinal side effects persisted, flutamide was discontinued. The protocol also mandated close monitoring of liver enzyme levels. If one or more level rose to above 1.5 times the top of the normal range, dose reduction of flutamide was recommended. If the level(s) continued to rise to above twice the top of the normal range, discontinuation was recommended.

Radiation treatment was delivered to a target volume that included the whole of the prostate (and seminal vesicles/base of bladder if clinically involved) plus a margin of 15 mm in all dimensions. Pelvic lymph nodes were not included in the target volume for any part of the treatment. The planning target volume (PTV) was defined using a planning CT scan, with contouring of all slices through the length of the PTV. Investigators used additional localisation techniques, such as a urethrogram or gastro-intestinal contrast, at their discretion. The technique used for delivery of the radiation treatment was at the discretion of the investigator, but it had to include at least three fields, and the use of shielding to limit the dose to tissue outside the target volume was encouraged. The total dose to the target volume was set at 66.00 Gy, prescribed according to ICRU 50, delivered in 33 fractions over 45 days. A technical audit on protocol compliance was performed, and has already been reported [20].

In all patients, an estimate of the size of the radiation target volume was made by multiplication of the *x*, *y* and *z* dimensions (in centimetres) of the radiation fields used. The use of shielding was noted, but no algorithm was available to calculate the precise reduction in target volume achieved by shielding in individual cases. The median radiation target volume for each of the treatment arms was compared, to see whether MAD prior to radiotherapy led to a reduction in the target volume, and to see whether such an effect was dependent on the duration of MAD. In addition, analysis was performed to see if there was correlation between the target volume size and the severity of acute urinary and bowel side effects of radiation treatment.

The immediate side effects of MAD and radiation treatment were quantified using a self-assessment questionnaire (SAQ), which comprehensively assesses bowel and urinary symptoms, and sexual function, in the period immediately prior to completion of the questionnaire. The investigator, or his research staff, gave the SAQ to the patient

to complete at specified times before, during, and after treatment. To maintain confidentiality, the SAQ was coded with the patient's trial number, and he was encouraged to complete the SAQ unassisted either in the clinic or in the privacy of his own home.

The SAQ was designed specifically for Australian and New Zealand patients, and although not validated as a whole, the various components had been previously tested. In addition to questions designed specifically to determine the EORTC/RTOG (and more recently CTC) grading of acute and delayed bowel and urinary toxicity, the SAQ included questions addressing bowel symptoms documented in previous TROG trials [11]. Questions addressing the levels of inconvenience caused by the patients' symptoms, as developed by Litwin et al. in the UCLA Quality of Life project, completed the questionnaire. The SAQ therefore enabled the patient to score key bowel and urinary symptoms, as well as the overall inconvenience these symptoms caused. Baseline scores for all symptoms were those recorded at registration. Assessment scores were recorded immediately before radiotherapy for the two MAD treatment arms, and at the end of radiotherapy for all patients. Changes in score from baseline to immediately prior to radiotherapy were used to determine the influence of MAD on bowel and urinary function, while changes in symptomatology from immediately prior to radiotherapy and the end of radiotherapy were used to determine changes in symptomatology attributable to radiotherapy in the three arms.

Sexual potency is a difficult parameter to assess in a population of older men, as factors other than erectile dysfunction are sometimes responsible for patients being sexually inactive. For this reason, the key information obtained from the SAQ was whether or not the patient was sexually active at a given time. These data allowed calculation of the proportion of sexually active patients in each treatment arm at baseline, and at various times during and after treatment. As impotence is a recognised side effect of both MAD and radiotherapy, but for MAD is believed to be temporary only, a comparison of the proportions of sexually active patients in each treatment arm at 12 months after completion of radiation treatment was made. This allowed the reversibility of MAD effects to be determined. Finally, analysis was performed to see if younger patients were less likely than older ones to experience sexual dysfunction as a result of treatment.

Due to the ordinal nature of the data, comparisons between groups of patients experiencing symptoms at or above a given grade were undertaken using the Mann–Whitney procedure. Probability levels below 0.05 were considered significant. Multiple logistical regression modelling procedures were used to determine the independence of associations between symptoms due to radiation treatment and various explanatory variables, such as treatment arm, age, radiation target volume, and symptom levels prior to any treatment (i.e. baseline values).

3. Results

3.1. Patient data

Out of a total of 818 patients, 800 from 18 centres across Australia and New Zealand were fully evaluable. There were no significant imbalances between the trial arms with respect to the stratification variables: age, stage, Gleason score and pre-treatment PSA level (Table 1).

Self-assessment questionnaires (SAQs) were completed in full by 769 (96.1%) of the 800 evaluable patients prior to any treatment (i.e. at baseline), 721 (90.1%) prior to radiotherapy, and 738 (92.3%) after radiotherapy. Missing questionnaire data at baseline and after radiotherapy were equally distributed between treatment arms, and did not relate to patient age or treatment centre. Patients on the two MAD arms were less likely to complete a SAQ prior to radiation treatment compared to patients receiving radiation treatment alone (when the baseline SAQ served this purpose). Investigation of the 38 missing or incomplete SAQs before radiation treatment in the MAD arms suggested that the most likely explanation was a failure to administer the questionnaire to patients living at a distance from the treatment centre, and that patient related factors did not play a part. These observations were supported by the finding that responses to the baseline and post-radiotherapy SAQs by these 38 patients were very similar in all respects to the responses given by patients who completed pre-radiotherapy questionnaires in these arms. Systematic failure or refusal to complete the SAQ was uncommon. Only three of 31 (9.7%) patients, who did not fully complete their baseline questionnaires, failed to complete subsequent questionnaires.

3.2. Pre-treatment (baseline) questionnaire responses

Pre-treatment questionnaires indicated that significant numbers of patients in all trial arms experienced, and were troubled by both bowel and urinary symptoms, without bias with respect to trial arm.

Table 2 shows that the main bowel symptoms prior to any treatment were frequency (8.5–10.0% patients

experienced bowel action frequency ‘greater than twice daily’), looseness (6.6–12.7% patients had passed loose motions ‘half the time’ in the previous 4 weeks), and urgency (3.1–4.9% patients felt they needed to defaecate ‘immediately’ upon the first urge to do so). Overall, 1.2–2.3% patients reported that their bowel symptoms were at least a moderate problem to them.

Table 3 shows that urinary symptoms prior to any treatment were more common. The main ones were poor stream (27.3–29.5% patients reported their urinary stream was ‘weak all the time’ or worse), frequency (45.5–52.5% patients required to micturate at intervals less than 3 h during the day, and 23.2–25.9% patients as frequently during the night), and dysuria (3.8–6.9% patients had discomfort ‘quite frequently but bearable’ or worse). Overall, 5.4–9.4% patients considered their urinary symptoms were at least a ‘moderate’ problem to them.

Table 4 shows that 33–38% patients indicated that they were ‘sexually active’ at the time of randomisation. As anticipated sexual activity was age dependent. Of the 88 patients under 60 years, 61.4% were sexually active. For patients between 60 and 70 years the percentage fell to 39.6%, and for patients over 70 years the percentage was just 24.2%.

3.3. Tolerance and compliance with maximal androgen deprivation prior to radiotherapy

Tables 2 and 3 summarise the bowel and urinary symptoms experienced by patients immediately prior to radiotherapy. Patients on the 3-month androgen deprivation arm would have received 2 months MAD at the time of the questionnaire, and patients on the 6-month arm, 5 months MAD.

The change in symptomatology attributable to MAD is shown in Figs. 2 and 3.

From Fig. 2 it may be seen that worsening of bowel symptoms occurred in both androgen deprivation arms with little difference between the two. Bowel frequency deteriorated by two grades in the 3- and 6-month androgen deprivation arms in 3.1 and 1.8% of patients, respectively, and motions became ‘more loose’ by two grades in 7.1 and

Table 1
Composition of the treatment arms by the main stratification variables

	XRT	3 Month	6 Months
Eligible	269	264	267
Ineligible	9	6	5
Age (Median (range))	67 (51–80)	68 (48–81)	68 (41–87)
Stage Nos (%)			
T2 B,C	163 (60.6)	154 (58.3)	162 (60.7)
T3,4	106 (39.4)	110 (41.7)	105 (39.3)
Gleason score (Median (range))	7 (2–10)	7 (2–10)	7 (2–9)
Initial PSA	16.35 (0.5–165)	14.4 (0.5–154.2)	14.5 (1.1–203.9)

Table 2
Bowel symptom data

	Baseline data						Pre radiotherapy data				Post radiotherapy data					
	Radiotherapy <i>N</i> = 269	(%)	3 Months MAD <i>N</i> = 264	(%)	6 Months MAD <i>N</i> = 267	(%)	3 Months MAD <i>N</i> = 264	(%)	6 Months MAD <i>N</i> = 267	(%)	Radiotherapy <i>N</i> = 269	(%)	3 Months MAD <i>N</i> = 264	(%)	6 Months MAD <i>N</i> = 267	(%)
Total randomised																
Bowel frequency																
Total respondents	264	(98.1%)	259	(98.1%)	259	(97.0%)	229	(86.7%)	231	(86.5%)	253	(94.1%)	253	(95.8%)	247	(92.5%)
Number missing	5	(1.9%)	5	(1.9%)	8	(3.0%)	35	(13.3%)	36	(13.5%)	16	(5.9%)	11	(4.2%)	20	(7.5%)
‘More than twice daily’ ^a	24	(9.1%)	22	(8.5%)	26	(10.0%)	22	(9.6%)	23	(10.0%)	124	(49.0%)	121	(47.8%)	102	(41.3%)
Bowel looseness																
Total respondents	259	(96.3%)	257	(97.3%)	256	(95.9%)	231	(87.5%)	231	(86.2%)	252	(93.7%)	253	(95.8%)	247	(92.5%)
Number missing	10	(3.7%)	7	(2.7%)	11	(4.1%)	33	(12.5%)	37	(13.8%)	17	(6.3%)	11	(4.2%)	20	(7.5%)
‘About half the time or more’ ^a	33	(12.7%)	31	(12.1%)	17	(6.6%)	39	(16.9%)	34	(14.7%)	73	(29.0%)	76	(30.0%)	69	(27.9%)
Bowel pain																
Total respondents	264	(98.1%)	259	(98.1%)	259	(97.0%)	229	(86.7%)	232	(86.9%)	253	(94.1%)	253	(95.8%)	246	(92.1%)
Number missing	5	(1.9%)	5	(1.9%)	8	(3.0%)	35	(13.3%)	35	(13.1%)	16	(5.9%)	11	(4.2%)	21	(7.9%)
‘Moderate pain’ or worse on defaecation ^a	1	(0.4%)	3	(1.2%)	1	(0.4%)	0	(0.0%)	4	(1.7%)	64	(25.3%)	72	(28.5%)	68	(27.6%)
Urgency																
Total respondents	263	(97.8%)	259	(98.1%)	258	(96.6%)	229	(86.7%)	232	(86.9%)	253	(94.1%)	253	(95.8%)	246	(92.1%)
Number missing	6	(2.2%)	5	(1.9%)	9	(3.4%)	35	(13.3%)	35	(13.1%)	16	(5.9%)	11	(4.2%)	21	(7.9%)
‘Need to defaecate immediately’ on urge ^a	13	(4.9%)	9	(3.5%)	8	(3.1%)	7	(3.1%)	12	(5.2%)	59	(23.3%)	53	(20.9%)	63	(25.6%)
Rectal blood																
Total respondents	261	(97.0%)	256	(97.0%)	254	(95.1%)	232	(87.9%)	231	(86.5%)	252	(93.7%)	252	(95.5%)	247	(92.5%)
Number missing	8	(3.0%)	8	(3.0%)	13	(4.9%)	32	(12.1%)	36	(13.5%)	17	(6.3%)	12	(4.5%)	20	(7.5%)
‘Pass blood between motions’ or worse ^a	1	(0.4%)	1	(0.4%)	0	(0.0%)	1	(0.4%)	1	(0.4%)	2	(0.8%)	8	(3.2%)	8	(3.2%)
Inconvenience																
Total respondents	259	(96.3%)	257	(97.3%)	257	(96.3%)	232	(87.9%)	230	(86.1%)	251	(93.3%)	253	(95.8%)	244	(91.4%)
Number missing	10	(3.7%)	7	(2.7%)	10	(3.7%)	32	(12.1%)	37	(13.9%)	18	(6.7%)	11	(4.2%)	23	(8.6%)
‘Moderate problem’ or worse ^a	6	(2.3%)	5	(1.9%)	3	(1.2%)	6	(2.6%)	8	(3.5%)	64	(25.5%)	59	(23.3%)	57	(23.4%)
‘No problem at all’ ^a	209	(80.7%)	214	(83.3%)	217	(84.4%)	180	(77.6%)	185	(80.4%)	59	(23.5%)	72	(28.5%)	74	(30.3%)

^a Percentage cited is percentage of total respondents not total randomised.

Table 3
Urinary symptom data

Urinary symptom data	Baseline data						Pre radiotherapy data						Post radiotherapy data					
	Radiotherapy N = 269	(%)	3 Months MAD N = 264	(%)	6 Months MAD N = 267	(%)	3 Months MAD N = 264	(%)	6 Months MAD N = 267	(%)	Radiotherapy N = 269	(%)	3 Months MAD N = 264	(%)	6 Months MAD N = 267	(%)		
Daytime frequency																		
Total respondents	263	(97.8%)	257	(97.3%)	261	(97.8%)	231	(87.5%)	231	(86.5%)	248	(92.2%)	253	(95.8%)	247	(92.5%)		
Number missing	6	(2.2%)	7	(2.7%)	6	(2.2%)	33	(12.5%)	36	(13.5%)	21	(7.8%)	11	(4.2%)	20	(7.5%)		
'At intervals less than 3 h' ^a	138	(52.5%)	117	(45.5%)	127	(48.7%)	125	(54.1%)	94	(40.7%)	181	(73.0%)	187	(73.9%)	189	(76.5%)		
Nighttime frequency																		
Total respondents	263	(97.8%)	259	(98.1%)	260	(97.4%)	230	(90.6%)	231	(86.5%)	249	(92.6%)	254	(96.2%)	247	(92.5%)		
Number missing	6	(2.2%)	5	(1.9%)	7	(2.6%)	24	(9.4%)	36	(13.5%)	20	(7.4%)	10	(3.8%)	20	(7.5%)		
'At intervals less than 3 h' ^a	61	(23.2%)	67	(25.9%)	63	(24.2%)	74	(32.2%)	86	(37.2%)	140	(56.2%)	174	(68.5%)	173	(70.0%)		
Urinary pain																		
Total respondents	262	(97.4%)	259	(98.1%)	261	(97.8%)	231	(87.5%)	232	(86.9%)	250	(92.9%)	253	(95.8%)	247	(92.5%)		
Number missing	7	(2.6%)	5	(1.9%)	6	(2.2%)	33	(12.5%)	35	(13.1%)	19	(7.1%)	11	(4.2%)	20	(7.5%)		
'Quite frequently' or worse ^a	18	(6.9%)	12	(4.6%)	10	(3.8%)	6	(2.6%)	8	(3.4%)	81	(32.4%)	74	(29.2%)	71	(28.7%)		
Urinary stream																		
Total respondents	260	(96.7%)	257	(97.3%)	261	(97.8%)	231	(87.5%)	232	(86.9%)	250	(92.9%)	253	(95.8%)	247	(92.5%)		
Number missing	9	(3.3%)	7	(2.7%)	6	(2.2%)	33	(12.5%)	35	(13.1%)	19	(7.1%)	11	(4.2%)	20	(7.5%)		
Weak all the time or worse ^a	71	(27.3%)	68	(26.5%)	77	(29.5%)	48	(20.8%)	51	(22.0%)	106	(42.4%)	116	(45.8%)	121	(49.0%)		
Inconvenience																		
Total respondents	261	(97.0%)	256	(97.0%)	258	(96.6%)	231	(87.5%)	229	(85.8%)	251	(93.3%)	244	(92.4%)	243	(91.0%)		
Number missing	8	(3.0%)	8	(3.0%)	9	(3.4%)	33	(12.5%)	38	(14.2%)	18	(6.7%)	20	(7.6%)	24	(9.0%)		
'Moderate problem' or worse ^a	14	(5.4%)	24	(9.4%)	16	(6.2%)	9	(3.9%)	9	(3.9%)	44	(17.5%)	53	(21.7%)	52	(21.4%)		
'No problem at all' ^a	136	(52.1%)	147	(57.4%)	159	(61.6%)	145	(62.8%)	167	(72.9%)	71	(28.3%)	82	(33.6%)	80	(32.9%)		

^a Percentage cited is percentage of total respondents not total randomised.

Table 4
Reporting of sexual activity before, immediately after radiotherapy and 1 year after by treatment arm and age

	Radiotherapy (n = 269)				3 Months MAD (n = 264)				6 Months MAD (n = 267)			
	Active	Inactive	Missing		Active	Inactive	Missing		Active	Inactive	Missing	
Sexual activity at baseline	101 37.5%	168 62.5%			88 33.3%	176 66.7%			96 36.0%	171 64.0%		
Sexual activity—post radiotherapy	50 18.6%	209 77.7%	10 3.7%		8 3.0%	254 96.2%	2 0.8%		7 2.6%	260 97.4%	2 0.8%	
Sexual activity—1 year post radiotherapy	50 18.6%	217 80.7%	2 0.7%		32 12.1%	230 87.1%	2 0.8%		38 14.2%	228 85.4%	1 0.4%	
	Under 60 (n = 88)				60–70 (n = 386)				Over 70 (n = 326)			
Sexual activity at baseline	Active 54 61.4%	Inactive 34 38.6%	Missing 0.0%		Active 152 39.6%	Inactive 232 60.4%	Missing 0.0%		Active 79 24.2%	Inactive 247 75.8%	Missing 0.0%	
Sexual activity—post radiotherapy	13 14.8%	75 85.2%	0.0%		35 9.1%	343 88.9%	8 2.1%		17 5.2%	305 93.6%	4 1.2%	
Sexual activity—1 year post radiotherapy	29 33.0%	59 67.0%	0.0%		60 15.5%	321 83.2%	5 1.3%		31 9.5%	295 90.5%	0.0%	

9.0% of patients. Overall, this increase in bowel activity was reported to represent a worsening in perceived bowel problems of at least two grades in 4.4% of patients treated with 3 months MAD, and 7.6% of patients treated with 6 months MAD.

Fig. 3 indicates that patients often experienced improvement in urinary frequency, pain and stream, rather than deterioration. Daytime frequency was reported to improve significantly more often in patients treated on the 6-month arm ($P < 0.001$), whereas night-time frequency worsened more often on this arm ($P < 0.02$). Overall, there were more patients who perceived that MAD led to an improvement in urinary function, rather than patients who felt it led to deterioration. The percentage of patients reporting no problems with micturition prior to radiation treatment increased from 62.8% in the 3-month arm to 72.9% ($P < 0.02$) in the 6-month arm. Only 52.1% of patients treated on the radiotherapy alone arm were free of problems prior to radiation treatment.

Table 5 shows compliance with MAD. Derangement in liver function due to flutamide was common, and alanine aminotransferase and aspartate aminotransferase became elevated to a level 1.5 times greater than the upper limit of normal in up to 39% and 32% of patients, respectively. Elevation to level(s) more than twice normal contributed to discontinuation of flutamide in 13.6% and 16.6% of patients in the two MAD arms. Unwanted bowel effects contributed to discontinuation in 3.8% and 7.9% of patients. Overall, there were slightly more discontinuations on the 6-month treatment arm, compared to the 3-month arm (26.9% vs. 20.1%), but the difference was not statistically significant. Goserelin was discontinued in just 10 patients in both the MAD treatment arms. Stopping was for a variety of reasons, and apparently occurred in error on four occasions.

3.4. Impact of androgen deprivation on target volume size choices

Table 6 shows that target volume choices were influenced by duration of MAD. The patients treated on both androgen deprivation arms had significantly smaller volumes than those treated by radiotherapy alone ($P < 0.001$, respectively). The difference between the two androgen deprivation arms did not reach significance ($P = 0.06$). It must be noted, however, that no prostate volume data per se were available to confirm that target volume choice was based exclusively on prostate volume.

3.5. Bowel and bladder symptoms attributable to radiotherapy

Despite smaller target volumes being treated in the two androgen deprivation arms, bowel symptoms, and the inconvenience caused by them, were similar at the end of radiation treatment in all treatment arms (Table 2), and this was also true for urinary symptoms and inconvenience

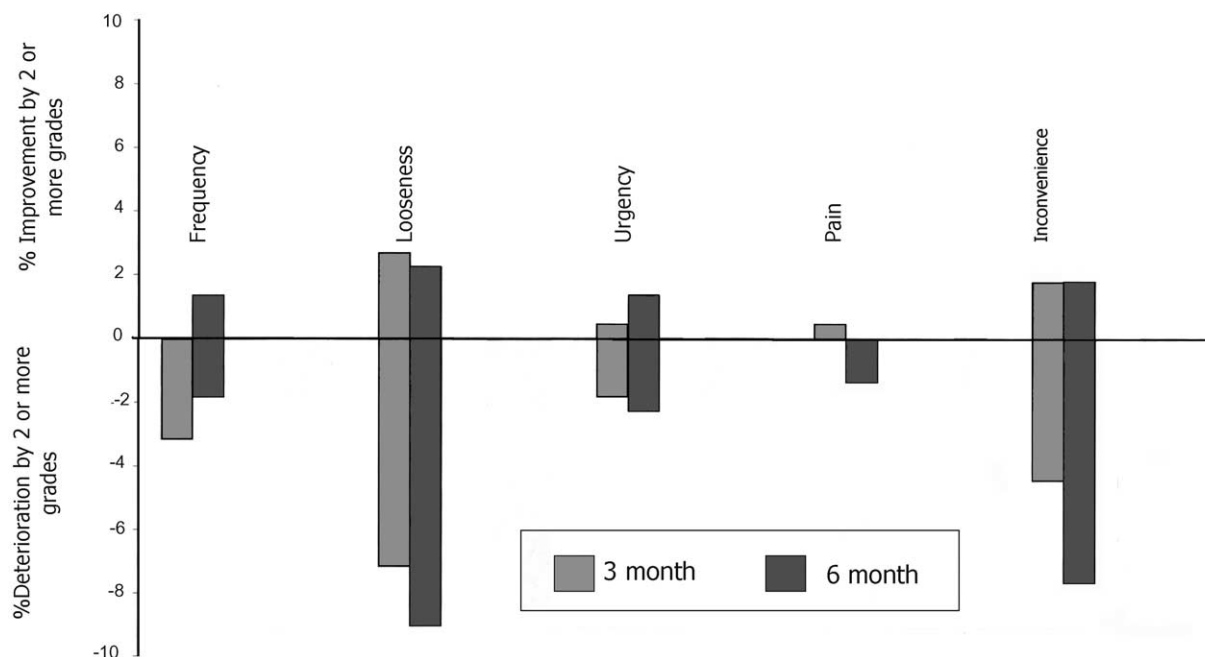


Fig. 2. Change in bowel symptomatology during androgen deprivation (baseline vs. start RT).

(Table 3). Change in bowel symptomatology from beginning to end of radiation treatment is shown in Fig. 4. MAD had very limited impact on bowel symptoms due to radiation treatment. Change in urinary symptomatology from beginning to end of radiation treatment is shown in Fig. 5. It can be seen that the improvements in urinary symptomatology, which occurred in the MAD arms prior to radiation treatment (see Fig. 3) were offset by significantly more urinary symptoms and inconvenience occurring in these patients as a result of radiation treatment.

In uni-variate analyses, the severity of all bowel and urinary symptoms reported at the end of radiotherapy were found to correlate with symptom levels at the time of randomisation (all associations $P < 0.0001$). All of these associations were confirmed in multi-variate analyses, which also showed that severity of symptoms was independent of age, stage, treatment arm and target volume size. In these models, the only other factors to exhibit independent association with symptoms at the end of radiotherapy were increasing age, which was predictive of

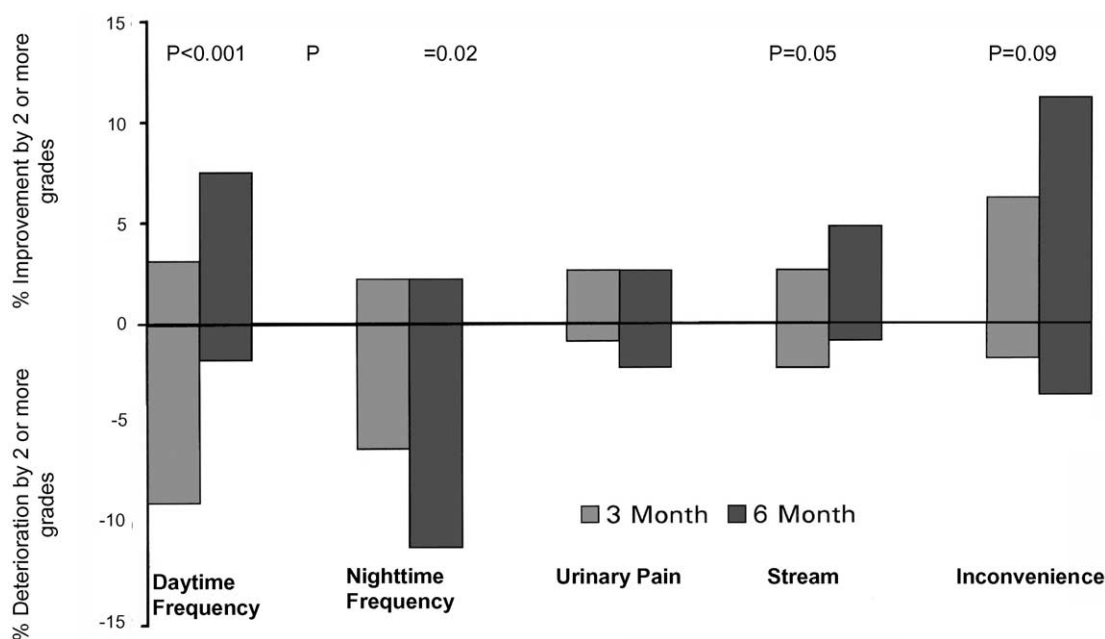


Fig. 3. Change in urinary symptomatology during androgen deprivation (baseline vs. start RT).

Table 5
Tolerance to and compliance with androgen deprivation medication

	3 Months MAD		6 Months MAD	
	Count	Percent	Count	Percent
Total commencing flutamide	264		267	
Total discontinuing flutamide	53	(20.1%)*	71	(26.9%)*
Reasons for discontinuing flutamide				
Bowel effect only	8	(3.0%)	13	(4.9%)
Bowel effect and LFT toxicity	1	(0.4%)	5	(1.9%)
Bowel effect and other reasons	1	(0.4%)	3	(1.1%)
LFT toxicity only	27	(10.2%)	36	(13.6%)
LFT toxicity and other reasons	8	(3.0%)	3	(1.1%)
Other reasons	9	(3.4%)	14	(5.3%)
Total commencing Goserelin	264	267		
Reasons for discontinuing Goserelin				
General malaise, intolerable hot flushes	1	(0.4%)		
Worsening of Parkinsonism, focal myalgia			1	(0.4%)
Aversion to injections, hot flushes			1	(0.4%)
Severe depressive illness			1	(0.4%)
Severe headaches			1	(0.4%)
Rash, aversion to injections			1	(0.4%)
Stopped in error	1	(0.4%)	3	(1.1%)

LFT, liver function tests; * $P = 0.065$.

lower severities of bowel pain, frequency and bleeding, and urinary stream difficulty, and androgen deprivation, which was predictive of greater nocturnal urinary frequency.

3.6. Sexual activity after radiotherapy

As indicated in Table 4 (presented in the bar graphs of Figs. 6 and 7), just 3% of patients on the two androgen deprivation arms were sexually active at the end of radiation treatment. The corresponding percentage of those treated on the radiation treatment only arm was 19%. At 1 year post treatment, however, the percentage of patients who reported sexual activity was similar in all three trial arms (12.1–18.6%). Patient age was observed to be a major factor in predicting the probability of sexual activity at 1 year after radiation treatment. At that time, 33% patients under 60 years, 15.5% patients 60–70 years, and 9.5% patients over 70 years were sexually active. Of patients sexually active at randomisation, those under 60 years were more likely to be sexually active at 1 year (54%) compared

to those 60 years and over (39%), although the difference was not statistically significant.

4. Discussion

The over 90% return rate for completed self-assessment questionnaires reflects well on the enthusiasm of the trial clinicians and their research staff. Indeed, this rate of return is at the upper end of the range for such data in randomised controlled trials [2]. However, the credibility of conclusions drawn from self-assessment data is not only dependent on questionnaire quality and completion rate. Efforts also need to be made to establish whether incomplete or missing data were associated with specific sub-groups whose responses, had they been known, could have altered the conclusions drawn. We looked carefully at this possibility in our analysis, and having found no reason to suspect biases, feel that the views of the 90% or more who did respond are representative of the entire trial population.

It is interesting to note that many patients had urinary and bowel symptoms and were troubled by them before any treatment for their prostate cancer. This reflects the fact that such symptoms are prevalent in a male population with a median age of 68 years. Sexual inactivity is also common in this age group, and it needs to be recognised that this state is often physiological, and not a consequence of treatment.

The impact of acute side effects of neo-adjuvant MAD are not well documented. The liver and bowel toxicity of flutamide has been previously reported [13], but it was hoped that these side effects would have relatively small impact on patient compliance when the duration of treatment was short. In fact, this did not turn out to be

Table 6
Impact of androgen deprivation on choice of target volume

	Treatment arm		
	Radiotherapy	3 Months MAD	6 Months MAD
Number of observations	267	264	264
Median volume	924	810	765
Range	394–2050	394–2511	333–2156
Geometric mean	900 ¹	815 ²	772 ³

P -values 1 vs. 2 $P < 0.001$; 2 vs. 3 $P = 0.06$; 1 vs. 3 $P < 0.001$.

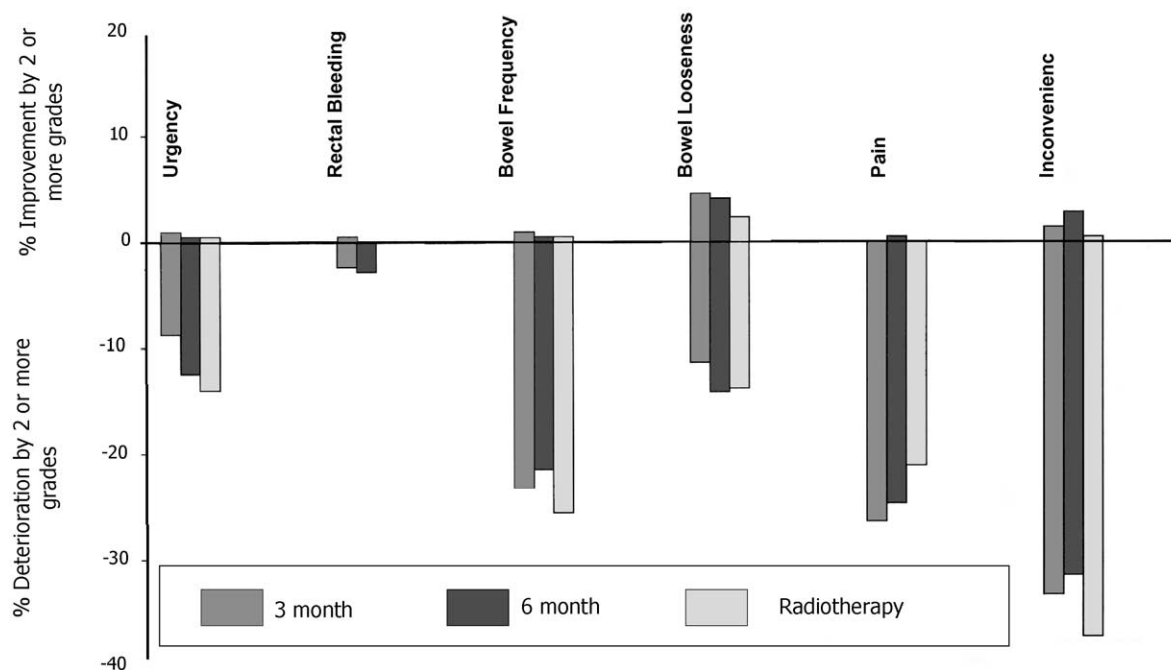


Fig. 4. Changes in bowel symptomatology during radiotherapy (start vs. end RT).

the case, as even with 3 months of planned treatment, 20% patients stopped treatment early, and this figure rose to 27% when 6 months of treatment was planned. Analysis of the reasons for stopping showed that altered liver function was the most common reason, and this had usually developed by the end of the second month of treatment. Patients stopping

thereafter were more likely to do so because of bowel side effects. Our results suggest that these did not increase substantially in severity with more protracted treatment, but that patients were only prepared to tolerate bowel disturbance for a limited period. Based on these results, flutamide would have limitations as a long-term adjuvant

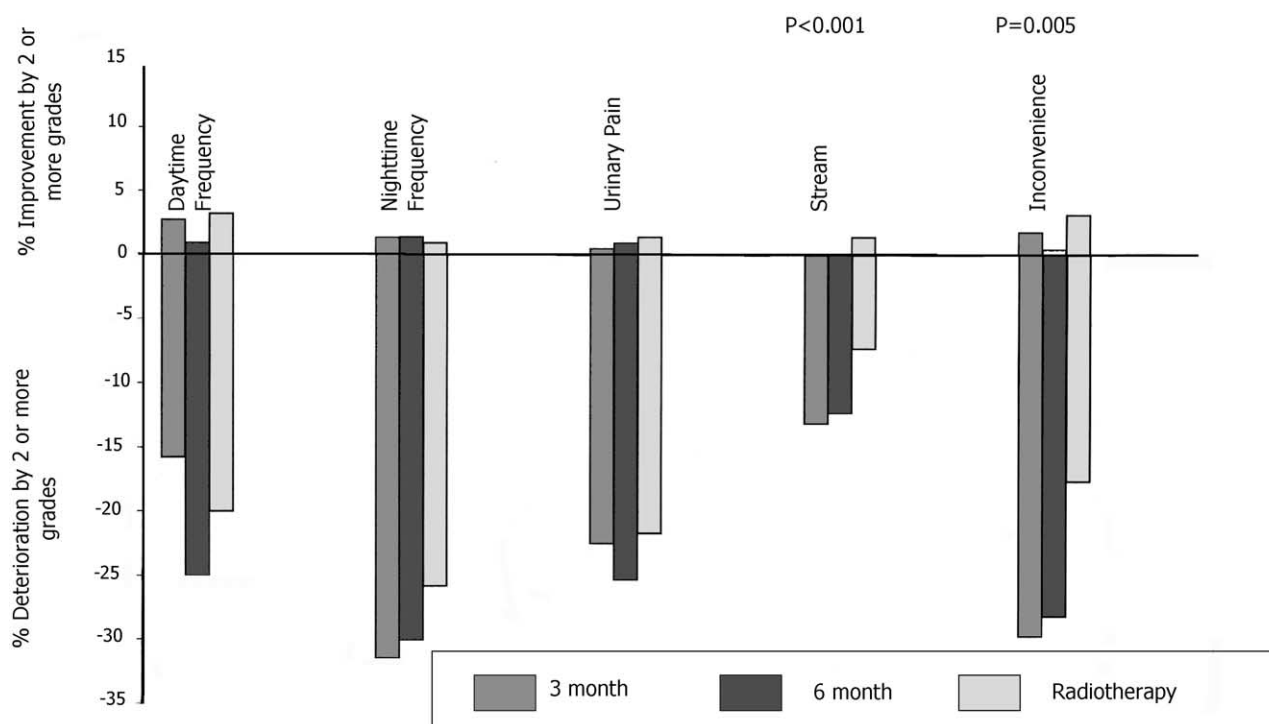


Fig. 5. Changes in urinary symptomatology during radiotherapy (start vs. end RT).

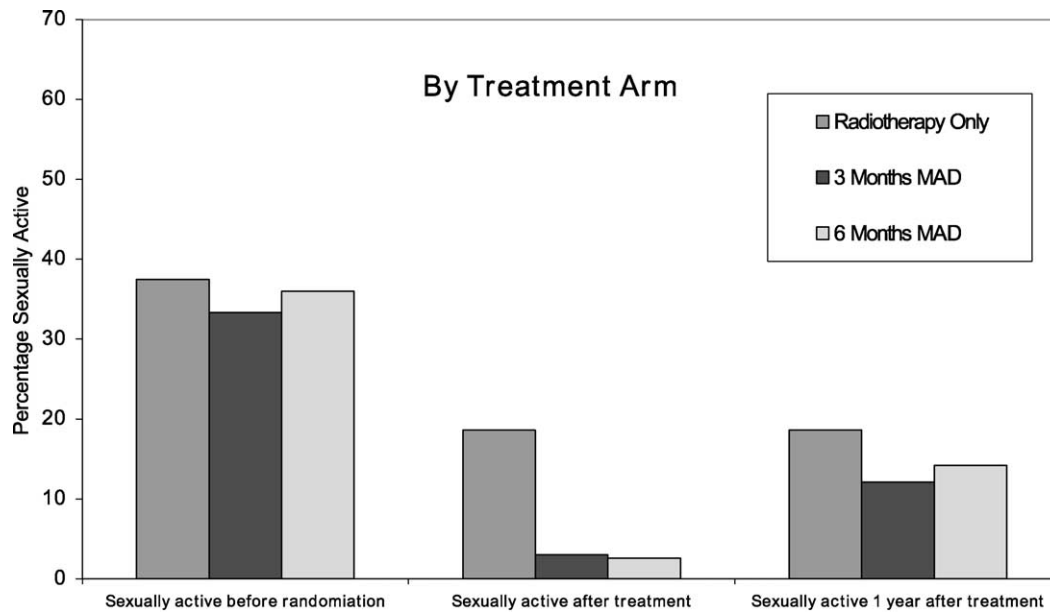


Fig. 6. Sexual activity prior to any treatment, immediately after radiotherapy and 1 year later.

hormone treatment. However, the newer anti-androgen bicalutamide is better tolerated [14], so this class of drug still has a potential role.

In contrast to flutamide, patient compliance with goserelin was excellent. Although most patients experienced hot flushes, these were rarely more than a minor irritation. The once monthly administration of the drug is likely to have facilitated the high compliance, so the current availability of 3 monthly depot injections makes LH–RH analogues all the more attractive for achieving longer term adjuvant androgen deprivation. In fact, published trials of

long term use of this class of drug suggest good compliance in this situation, although unwanted side effects are less well documented [3,4,7,12]. The biochemical changes induced by more long term use of LH–RH analogues may not always be reversible [15], and there are likely to be metabolic consequences, including loss of bone density, which result from protracted androgen deprivation [19,21,23]. Therefore, clinical trials investigating the benefits of adjuvant androgen deprivation need to incorporate methods that allow these unwanted effects to be quantified.

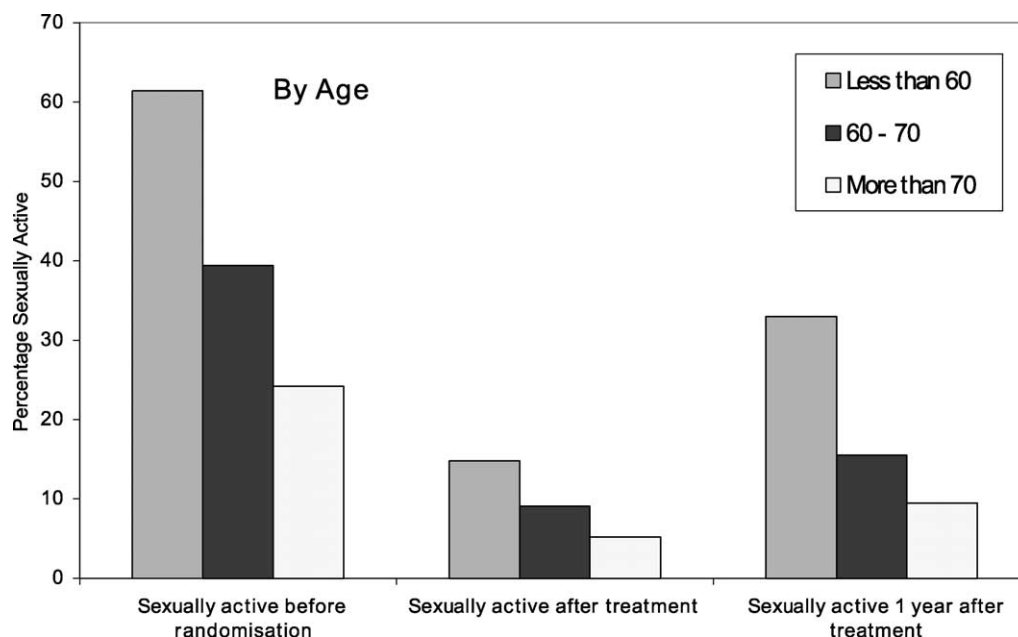


Fig. 7. Sexual activity prior to any treatment, immediately after radiotherapy and 1 year later.

Although the bowel side effects of flutamide have been previously reported, the effects of neo-adjuvant MAD on urinary function have not. MAD led to deterioration in night time frequency, but improvement in most other urinary symptoms. Overall, the percentage of patients with perceived micturition problems correlated inversely with the duration of MAD. This is an important finding, as these symptomatic benefits of MAD helped compensate for the side effects of this treatment. As a result, MAD was not perceived to cause major inconvenience.

Interestingly, the improvement in urinary symptomatology due to MAD was lost during the subsequent radiation treatment. At the end of radiation treatment, urinary symptoms, and the trouble caused by them were similar in all treatment arms. This was also true for bowel symptoms and the inconvenience these caused. The incidence and severity of radiation bowel and urinary acute side effects in our patients were similar to those in other published series [18]. There was certainly no evidence that the acute radiation side effects were exacerbated by neo-adjuvant MAD, and a recent publication [25] supports that there is no such interaction.

We were able to confirm other reports [9,16,24] that neo-adjuvant MAD does significantly reduce the target volume for radiotherapy, although our data did not demonstrate a statistically significant difference between 3- and 6-month MAD. However, it should be recognised that the tool used to measure the target volume in this trial is likely to have been too crude to detect more than gross differences. The target volume was calculated by multiplication of the *x*, *y* and *z* dimensions of the fields used to deliver treatment, and this does not take into account the large amount of shielding used by some investigators, especially those with access to multi-leaf collimation. Androgen deprivation has been reported to lead to continued shrinkage of the prostate to a minimum size at 6–7 months [6], but the absence of prostatic volume data in this trial makes it impossible for us to confirm this effect. The absence of prostatic volume data also makes it difficult to be sure that the reduction in target volume we have demonstrated was due to shrinkage of the prostate, or due to physician choice.

The positive correlation between size of radiation target volume size and incidence of late radiation bowel side effects is well known [5,10,16]. A similar correlation has been reported between target volume size and acute bowel and urinary side effects [8], but our analysis was unable to confirm this. In fact, multi-variate analysis demonstrated that the only factors to predict for severity of acute radiation side effects was the presence of bowel and/or urinary symptoms before any treatment, and age. It is well recognised that responses to self-assessment questionnaires reflect not only the severity of underlying pathological processes, but also patient consciousness of bodily function, fears of incapacity, and varying perceptions of the severity of symptomatology and the resulting inconvenience caused. Therefore, one interpretation of our analysis is that patients

who find variations in bodily habit inconvenient, and report them, are more likely to report in their questionnaires greater degrees of pre-treatment and post-treatment symptoms. If this is true, then our data suggest that younger patients are more likely to report inconveniencing symptoms than older ones. Without objective measures of urinary and bowel function in our patient population, it is not possible to be sure of the correct interpretation of these interesting associations.

The self-assessment questionnaire revealed that 33–38% of all patients (median age 68 years) were sexually active before any treatment was initiated for their prostate cancer. This trial has provided valuable information on the ability of patients to maintain sexual activity during treatment. Not surprisingly, only 3% patients were able to maintain sexual activity whilst receiving MAD as well as radiation treatment, whereas 19% patients receiving radiation treatment alone remained active. Interestingly, the percentage of patients sexually active 12 months after completion of treatment was similar (12.1–18.6%) in all three treatment arms. This implies that, following 3 and 6 months MAD, testosterone levels recover sufficiently in most patients to allow resumption of sexual activity. Our results also indicate that the radiation treatment used in this trial is responsible for over half of patients becoming sexually inactive by 12 months after treatment. This proportion can be expected to increase further with time, as erectile dysfunction due to radiation treatment takes up to 5 years to develop [17].

Analysis of sexual activity according to patient age did not produce any surprises. Younger patients were more likely to be active before any treatment, but the proportional reduction in activity due to treatment was similar in all age groups. However, our results suggest that recovery may be better in younger patients. Patients less than 60 years had a 54% chance of regaining sexually activity by 12 months after treatment, compared to 39% for patients aged 60 years and more. The fact that older patients have a relatively small chance of being sexually active 12 months after radiation treatment alone may assist them in determining the acceptability of undergoing adjuvant androgen deprivation as well, especially when this treatment is extended for longer periods than used in this trial.

5. Conclusion

Despite temporary effects on sexual activity, and compliance difficulties with flutamide, short-term neo-adjuvant MAD was not perceived by the patient to be a major inconvenience. If such MAD can be demonstrated to lead to improved biochemical control and/or survival, then patients would view these therapeutic gains as worthwhile. Compliance with short-term goserelin was excellent, suggesting that LH–RH analogues have a potential role in more protracted adjuvant treatment. However, in this

situation, survival advantages and side effects, especially metabolic, need to be assessed in parallel, in order to determine the optimal duration of treatment.

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